

Physics 1502Q: Equation Sheet for Exam 1

Physical Constants:

$$\begin{aligned}e &= 1.60 \times 10^{-19} \text{ C} && \text{(Fundamental unit of charge)} \\m_p &= 1.67 \times 10^{-27} \text{ kg} && \text{(Mass of a proton)} \\m_e &= 9.11 \times 10^{-31} \text{ kg} && \text{(Mass of an electron)} \\\epsilon_0 &= 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2 && \text{(Permittivity of free space)} \\k &= \frac{1}{4\pi\epsilon_0} \simeq 9.00 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2 && \text{(Coulomb constant)}\end{aligned}$$

Common SI Prefixes:

$$\begin{aligned}1 \text{ p} &= 1 \times 10^{-12} && \text{(pico)} \\1 \text{ n} &= 1 \times 10^{-9} && \text{(nano)} \\1 \text{ }\mu &= 1 \times 10^{-6} && \text{(micro)} \\1 \text{ m} &= 1 \times 10^{-3} && \text{(milli)} \\1 \text{ c} &= 1 \times 10^{-2} && \text{(centi)} \\1 \text{ k} &= 1 \times 10^3 && \text{(kilo)} \\1 \text{ M} &= 1 \times 10^6 && \text{(Mega)}\end{aligned}$$

Coulomb's Law: $\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$ (Coulomb force)

Electric Field: $\vec{E} = \frac{\vec{F}}{q_0}$ (General definition)

$$\vec{E} = k \frac{q}{r^2} \hat{r} \quad \text{(Point charge)}$$

$$d\vec{E} = k \frac{dq}{r^2} \hat{r} \quad \text{(Infinitesimal electric field)}$$

$$E = \frac{\sigma}{2\epsilon_0} \quad \text{(Electric field near infinite plane)}$$

Electric Flux: $\Phi_E = \int_S \vec{E} \cdot d\vec{A}$ (Electric flux through a surface S)

Gauss's Law: $\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$ (Net Electric flux through a closed surface S)

Charge Density: $\lambda = \frac{Q}{L}$ (Linear); $\sigma = \frac{Q}{A}$ (Surface); $\rho = \frac{Q}{V}$ (Volume)

Electrostatic Potential Energy:

$$\Delta U = -W \quad \text{(Potential Energy equals the negative of the work done BY a conservative force)}$$

$$\Delta U = - \int \vec{F} \cdot d\vec{s} \quad \text{(General Definition of Potential Energy)}$$

$$U = k \frac{q_1 q_2}{r} \quad \text{(Potential energy of 2 point-charges relative to } \infty \text{)}$$

$$K = \frac{1}{2} m v^2 \quad \text{(Kinetic energy)}$$

$$K_f + U_f = K_i + U_i \quad \text{(Conservation of energy)}$$

Electric Potential:

$$\Delta V = \frac{\Delta U}{q} \quad (\text{General Definition of Electric Potential Difference})$$

$$\Delta V = V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s} \quad (\text{Electric Potential Difference from Electric Field})$$

$$V = \frac{kq}{r} \quad (\text{Electric Potential due to a point charge, relative to } \infty)$$

Some Geometry Equations: $A = 4\pi r^2$ (Surface area of a sphere)

$$A = 2\pi rL + 2\pi r^2 \quad (\text{Surface area of a cylinder})$$

$$V = \frac{4}{3}\pi r^3 \quad (\text{Volume of a sphere})$$

$$V = \pi r^2 L \quad (\text{Volume of a cylinder})$$